

Mathematical Expectation (Expected Value of a Random Variable) (12)

The 'average' value of a random phenomenon is termed as mathematical expectation or expected value.

Once we have constructed the probability distribution for a random variable, we often want to compute the mean or expected value of the random variable. The expected value of a discrete random variable is a weighted average of all possible values of the random variable, where the weights are the probabilities associated with the corresponding values.

The mathematical expression for computing the expected value of a discrete random variable X with probability mass function $f(x)$ is given below:

$$E(X) = \sum_x x \cdot f(x), \text{ (for discrete r.v.)}$$

The mathematical expression for computing the expected value of a continuous random variable X with probability density function $f(x)$ is however as follows —

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) \cdot dx, \text{ (for continuous r.v.)}$$

Expected Value of Function of a Random Variable

Consider a r.v. X with pdf (pmf) $f(x)$ & distribution function $F(x)$. If $g(\cdot)$ is a function s.t. $g(X)$ is a r.v. & $E[g(X)]$ exists (i.e. is defined), then —

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f(x) \cdot dx \quad [\text{for cont}^n \text{ r.v.}]$$

$$E[g(X)] = \sum_x g(x) \cdot f(x) \quad [\text{for discrete r.v.}]$$

Particular Cases (i)

If we take $g(x) = x^r$, r being a positive integer

$$E(X^r) = \int_{-\infty}^{\infty} x^r \cdot f(x) \cdot dx$$

which is defined as μ_r' , the r th moment (about origin) of the probability distⁿ.

$$\text{Thus, } \mu_r' \text{ (about origin)} = E(X^r)$$

$$\mu_1' \text{ (about origin)} = E(X)$$

$$\& \mu_2' \text{ (about origin)} = E(X^2)$$

$$\text{Hence, Mean} = \bar{x} = \mu_1' \text{ (about origin)} = E(X)$$

$$\& \mu_2 = \mu_2' - \mu_1'^2 = E(X^2) - \{E(X)\}^2$$

$$(2) \text{ If } g(x) = [X - E(X)]^r$$

$$g(x) = (x - \bar{x})^r$$

$$\therefore E[X - E(X)]^r = \int_{-\infty}^{\infty} [x - E(x)]^r f(x) \cdot dx$$

$$= \int_{-\infty}^{\infty} (x - \bar{x})^r f(x) \cdot dx$$

which is μ_r , the r th moment about mean

In particular, if $r=2$ —

$$\mu_2 = E[X - E(X)]^2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) \cdot dx$$

$$\mu_2 = E[X - E(X)]^2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) \cdot dx$$

$$(3) \text{ Taking } g(x) = \text{constant} = c \text{ (say)}$$

$$E(c) = \int_{-\infty}^{\infty} c \cdot f(x) \cdot dx = c \int_{-\infty}^{\infty} f(x) \cdot dx = c$$

$$E(c) = c$$

Properties(1) Addition Theorem of ExpectationStatement If X & Y are r.v.s, then

$$E(X+Y) = E(X) + E(Y)$$

provided all the expectations exist.

Proof.

Generalisation - The mathematical expectation of the sum of ' n ' random variables is equal to the sum of their expectations, provided all the expectations exist. Symbolically, if $X_1, X_2, \dots, X_n$ are random variables then

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

(2) Multiplication Theorem of ExpectationStatement:- If X & Y are independent random variables, then

$$E(XY) = E(X) \cdot E(Y)$$

Proof.

Generalisation - The mathematical expectation of the product of a number of independent random variables is equal to the product of their expectations. Symbolically, if $X_1, X_2, \dots, X_n$ are ' n ' independent r.v.s then

$$E(X_1, X_2, \dots, X_n) = E(X_1) \cdot E(X_2) \cdot \dots \cdot E(X_n)$$

(3) If X is a random variable & ' a ' is constant, then

$$(i) E[a\psi(X)] = a E[\psi(X)]$$

$$(ii) E[\psi(X) + a] = E[\psi(X)] + a$$

where $\psi(X)$, a fⁿ of X , is a r.v. & all the expectations exist.Proof.

Property-1 (Proof)

Let X & Y be continuous r.v.'s with joint pdf $f_{XY}(x, y)$ & marginal pdf's $f_X(x)$ & $f_Y(y)$ respectively. Then by defⁿ —

$$E(X) = \int_{-\infty}^{\infty} x \cdot f_X(x) dx$$

$$\& E(Y) = \int_{-\infty}^{\infty} y \cdot f_Y(y) dy$$

$$E(X+Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f_{XY}(x, y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \cdot f_{XY}(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \cdot f_{XY}(x, y) dx dy$$

$$= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{XY}(x, y) dy \right] dx + \int_{-\infty}^{\infty} y \left[\int_{-\infty}^{\infty} f_{XY}(x, y) dx \right] dy$$

$$= \int_{-\infty}^{\infty} x \cdot f_X(x) dx + \int_{-\infty}^{\infty} y \cdot f_Y(y) dy$$

$$\boxed{E(X+Y) = E(X) + E(Y)}$$

Generalisation:- The mathematical expectation of the sum of n random variables is equal to the sum of their expectations, provided all the expectations exist. Symbolically, if $X_1, X_2, \dots, X_n$ are random variables then —

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n) \quad (*)$$

$$\text{or } E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i), \text{ if all the expectations exist}$$

for two r.v.'s X_1 & X_2 , we get —

$$E(X_1 + X_2) = E(X_1) + E(X_2)$$

$\Rightarrow$ is true for $n=2$

Proof

Let us suppose that (*) is true for $n=r$ (say) —

$$E\left(\sum_{i=1}^r X_i\right) = \sum_{i=1}^r E(X_i)$$

$$\begin{aligned} E\left(\sum_{i=1}^{r+1} X_i\right) &= E\left[\sum_{i=1}^r X_i + X_{r+1}\right] \\ &= E\left(\sum_{i=1}^r X_i\right) + E(X_{r+1}) \\ &= \sum_{i=1}^r E(X_i) + E(X_{r+1}) \\ &= \sum_{i=1}^{r+1} E(X_i) \end{aligned}$$

If (*) is true for $n=r$, it is also true for $n=r+1$.

But we have proved in above that (*) is true for $n=2$.

Hence, it is true for $n=2+1=3$
 $n=3+1=4$

So on. Hence by the principle of mathematical induction (*) is true for all positive integral values of n .

Property-2 (Proof)

$$\begin{aligned} \therefore E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_X(x) f_Y(y) dx dy \quad [\because X \& Y \text{ are independent}] \\ &= \int_{-\infty}^{\infty} x f_X(x) dx \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= E(X) E(Y), \text{ provided } X \& Y \text{ are independent} \end{aligned}$$

Generalisation ~~the mathematical expectation~~

$$E(X_1, X_2, \dots, X_n) = E(X_1) \cdot E(X_2) \dots E(X_n)$$

$$\text{i.e. } E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n E(X_i)$$

provided all the expectations exist.

Proof.

for two independent r.v.s X_1 & X_2 , we get —

$$E(X_1 X_2) = E(X_1) E(X_2)$$

$\Rightarrow$ it is true for $n=2$

Let us suppose it is true for $n=r$ —

$$E\left(\prod_{i=1}^r X_i\right) = \prod_{i=1}^r E(X_i)$$

$$\text{Thus, } E\left(\prod_{i=1}^{r+1} X_i\right) = E\left(\prod_{i=1}^r X_i \cdot X_{r+1}\right)$$

$$= E\left(\prod_{i=1}^r X_i\right) \cdot E(X_{r+1})$$

$$= \left[\prod_{i=1}^r E(X_i)\right] \cdot E(X_{r+1})$$

$$= \prod_{i=1}^{r+1} E(X_i)$$

Hence, it is true for $n=r$, it is also true for $n=r+1$.
Hence by the principle of mathematical induction, it is
true for all positive integral values of n .

Property-3 (Proof)

$$(i) E[a\psi(X)] = \int_{-\infty}^{\infty} a\psi(x) \cdot f(x) dx$$

$$= a \int_{-\infty}^{\infty} \psi(x) f(x) dx$$

$$= a E[\psi(X)]$$

$$(ii) E[\psi(X) + a] = \int_{-\infty}^{\infty} \{\psi(x) + a\} f(x) dx$$

$$= \int_{-\infty}^{\infty} \psi(x) f(x) dx + a \int_{-\infty}^{\infty} f(x) dx$$

$$\left(\because \int_{-\infty}^{\infty} f(x) dx = 1 \right)$$

$$= E[\psi(X)] + a$$

Property-4 (Proof)

By def, we have —

$$E(aX+b) = \int_{-\infty}^{\infty} (ax+b) f(x) dx$$

$$= a \int_{-\infty}^{\infty} x f(x) dx + b \int_{-\infty}^{\infty} f(x) dx$$

$$= aE(X) + b$$

If X is continuous r.v. s.t. $X \geq 0$, then

Property-5

$$E(X) = \int_{-\infty}^{\infty} x \cdot p(x) dx$$

$$= \int_0^{\infty} x \cdot p(x) dx$$

provided the expectation exist.

[$\because$ If $X \geq 0$, $p(x) = 0$ for $x < 0$]

Property-7 (Proof) Since $Y \leq X$, we have —

$$Y - X \leq 0$$

$$\Rightarrow X - Y \geq 0$$

$$\text{Hence, } E(X - Y) \geq 0$$

$$\Rightarrow E(X) - E(Y) \geq 0$$

$$\Rightarrow E(X) \geq E(Y)$$

Property-8 (Proof) Since $X \leq |X|$, we have by property-7

$$E(X) \leq E|X|$$

Again $-X \leq |X|$, we have by property 7

$$E(-X) \leq E|X|$$

$$\therefore -E(X) \leq E|X|$$

Property-9 (Proof) $\int_{-\infty}^{\infty} |x|^s dF(x) = \int_{-1}^1 |x|^s dF(x) + \int_{|x|>1} |x|^s dF(x)$

If $s < r$, then $|x|^s < |x|^r$, for $|x| > 1$

$$\therefore \int_{-\infty}^{\infty} |x|^s dF(x) \leq \int_{-1}^1 |x|^s dF(x) + \int_{|x|>1} |x|^r dF(x)$$

$$\leq \int_{-1}^1 dF(x) + \int_{|x|>1} |x|^r dF(x)$$

$$\therefore \text{for } -1 < x < 1, |x|^s < 1$$

$$\therefore \int_{-\infty}^{\infty} |x|^s dF(x) \leq 1 + E|X|^r < \infty$$

$$\text{Hence } E(X^s) \text{ exists, } \forall 1 \leq s \leq r$$

Property-10 (Proof)

Let $f_X(x)$ & $g_Y(y)$ be the marginal pdf of X & Y resp. Since X & Y are independent, their joint pdf $f_{XY}(x, y)$ is —

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y) \quad \text{--- (*)}$$

By defⁿ, for contⁿ r.v. —

$$E[h(X) \cdot k(Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x) k(y) f(x, y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x) k(y) f(x) \cdot g(y) dx dy \quad (\text{from *})$$

Since $E[h(X) k(Y)]$ exists, the integral on the right-hand side is absolutely convergent & hence by Fubini's theorem for integral functions, we can change the order of integration —

$$\begin{aligned} E[h(X) k(Y)] &= \left[\int_{-\infty}^{\infty} h(x) f(x) dx \right] \left[\int_{-\infty}^{\infty} k(y) g(y) dy \right] \\ &= E[h(X)] \cdot E[k(Y)] \end{aligned}$$

Properties of Variance

Statement If X is a random variable, then $V(aX+b) = a^2 V(X)$ where ' a ' & ' b ' are constants.

Proof.

Covariance

If X & Y are two random variables, then covariance between them is defined as —

$$\begin{aligned} \text{Cov}(X, Y) &= E[\{X - E(X)\} \{Y - E(Y)\}] \\ &= E[XY - XE(Y) - YE(X) + E(X)E(Y)] \\ &= E(XY) - E(Y)E(X) - E(X)E(Y) + E(X)E(Y) \end{aligned}$$

$$\boxed{\text{Cov}(X, Y) = E(XY) - E(X)E(Y)}$$

If X & Y are independent then $E(XY) = E(X)E(Y)$ & hence in this case

$$\boxed{\text{Cov}(X, Y) = E(X)E(Y) - E(X)E(Y) = 0}$$

Remark ① $\text{Cov}(aX, bY) = E[\{aX - E(aX)\} \{bY - E(bY)\}]$
 $= E[a\{X - E(X)\} b\{Y - E(Y)\}]$
 $= ab E[\{X - E(X)\} \{Y - E(Y)\}]$
 $= ab \text{Cov}(X, Y)$

② $\text{Cov}(X+a, Y+b) = \text{Cov}(X, Y)$

③ $\text{Cov}\left(\frac{X - \bar{X}}{\sigma_X}, \frac{Y - \bar{Y}}{\sigma_Y}\right) = \frac{1}{\sigma_X \sigma_Y} \text{Cov}(X, Y)$

Variance of a linear combination of Random Variable

$$V\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 V(X_i) + 2 \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)$$

Proof.

Proof of Variance

Let $Y = aX + b$. Then $E(Y) = aE(X) + b$

$$\therefore Y - E(Y) = a[X - E(X)]$$

Squaring & taking expectation on both sides, we get -

$$E[Y - E(Y)]^2 = a^2 E[X - E(X)]^2$$

$$\Rightarrow V(Y) = a^2 V(X)$$

$$\text{or } V(aX + b) = a^2 V(X)$$

Cor. (i) If $b=0$, then $V(aX) = a^2 V(X)$

$\Rightarrow$ Variance is not independent of change of scale.

(ii) If $a=0$, then $V(b)=0$

$\Rightarrow$ Variance of a constant is zero.

(iii) If $a=1$, then $V(X+b) = V(X)$

$\Rightarrow$ Variance is independent of change of origin.

Property-4 Similarly, we shall get:

$$\bullet \text{Cov}(aX+b, cY+d) = ac \text{Cov}(X, Y)$$

$$\bullet \text{Cov}(X+Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$$

$$\bullet \text{Cov}(aX+bY, cX+dY) = ac\sigma_X^2 + bd\sigma_Y^2 + (ad+bc) \text{Cov}(X, Y)$$

Property-5 If X & Y are independent, $\text{Cov}(X, Y) = 0$

However the converse is not true.

Variance of a Linear Combination of Random Variables

Let $X_1, X_2, \dots, X_n$ be n random variables, then

$$V\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 V(X_i) + 2 \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)$$

Proof - Let $U = a_1 X_1 + a_2 X_2 + \dots + a_n X_n$

so that, $E(U) = a_1 E(X_1) + a_2 E(X_2) + \dots + a_n E(X_n)$

$$\therefore U - E(U) = a_1 \{X_1 - E(X_1)\} + a_2 \{X_2 - E(X_2)\} + \dots + a_n \{X_n - E(X_n)\}$$

Squaring & taking expectation on both sides, we get -

$$E[U - E(U)]^2 = a_1^2 E\{X_1 - E(X_1)\}^2 + a_2^2 E\{X_2 - E(X_2)\}^2 + \dots + a_n^2 E\{X_n - E(X_n)\}^2 + 2 \sum_{i=1}^n \sum_{j=1}^n a_i a_j E\{X_i - E(X_i)\} \{X_j - E(X_j)\}$$

$$\Rightarrow V(U) = a_1^2 V(X_1) + a_2^2 V(X_2) + \dots + a_n^2 V(X_n) + 2 \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)$$

$$\Rightarrow V\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 V(X_i) + 2 \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)$$

Remark-① If $a_i = 1$; $i = 1, 2, \dots, n$ then

$$V(X_1 + X_2 + \dots + X_n) = V(X_1) + V(X_2) + \dots + V(X_n) + 2 \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j)$$
$$i < j$$

Moments

The r th moment of a variable x about any point $x=A$, ~~usually~~ usually denoted by μ_r' is given by —

$$\mu_r' = \frac{1}{N} \sum f_i (x_i - A)^r, \quad \sum f_i = N$$

$\mu_r' = \frac{1}{N} \sum f_i d_i^r$, where $d_i = x_i - A$
The r th moment of a variable x about the mean $\bar{x}$, usually denoted by μ_r is given by —

$$\mu_r = \frac{1}{N} \sum f_i (x_i - \bar{x})^r = \frac{1}{N} \sum f_i z_i^r \quad \text{where } z_i = x_i - \bar{x}$$

In particular,

$$\mu_0 = \frac{1}{N} \sum f_i (x_i - \bar{x})^0 = \frac{1}{N} \sum f_i = 1$$

$$\mu_1 = \frac{1}{N} \sum f_i (x_i - \bar{x}) = 0$$

Being the algebraic sum of deviations from the mean

$$\mu_2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2 = \sigma^2$$

These results, viz, $\mu_0=1$, $\mu_1=0$ & $\mu_2=\sigma^2$, are ~~the~~ of fundamental importance & should be committed to memory. We know that if

$d_i = x_i - A$, then

$$\bar{x} = A + \frac{1}{N} \sum f_i d_i = A + \mu_1' \quad (*)$$

Relation between Moments about Mean in Terms of Moments about Any point & Vice Versa

$$\mu_r = \frac{1}{N} \sum f_i (x_i - \bar{x})^r = \frac{1}{N} \sum f_i (x_i - A + A - \bar{x})^r = \frac{1}{N} \sum f_i (d_i + A - \bar{x})^r$$

where $d_i = x_i - A$

Using (*), we get —

$$\begin{aligned} \mu_r &= \frac{1}{N} \sum f_i (d_i - \mu_1')^r \\ &= \frac{1}{N} \sum f_i \left[d_i^r - r C_1 d_i^{r-1} \mu_1' + r C_2 d_i^{r-2} \mu_1'^2 - r C_3 d_i^{r-3} \mu_1'^3 + \dots + (-1)^r \mu_1'^r \right] \end{aligned}$$